



CS201 DSA PROJECT

Reimagine Sentiment Analysis for Airline Reviews Insights

Data Structures for
Polarized Word-Frequency Counts

G2T3 - TEAM TOUCH GRASS



NOV 2025



Problem Statement

**Information
Overload**



**Difficult to Interpret
Large Volumes of Text**

**Sentiment
Differentiation**



**Words Distinguish Positive
from Negative reviews?**

**Efficient
Computation at Scale**



**Slow and memory-heavy
as Sentiment Data Scales**

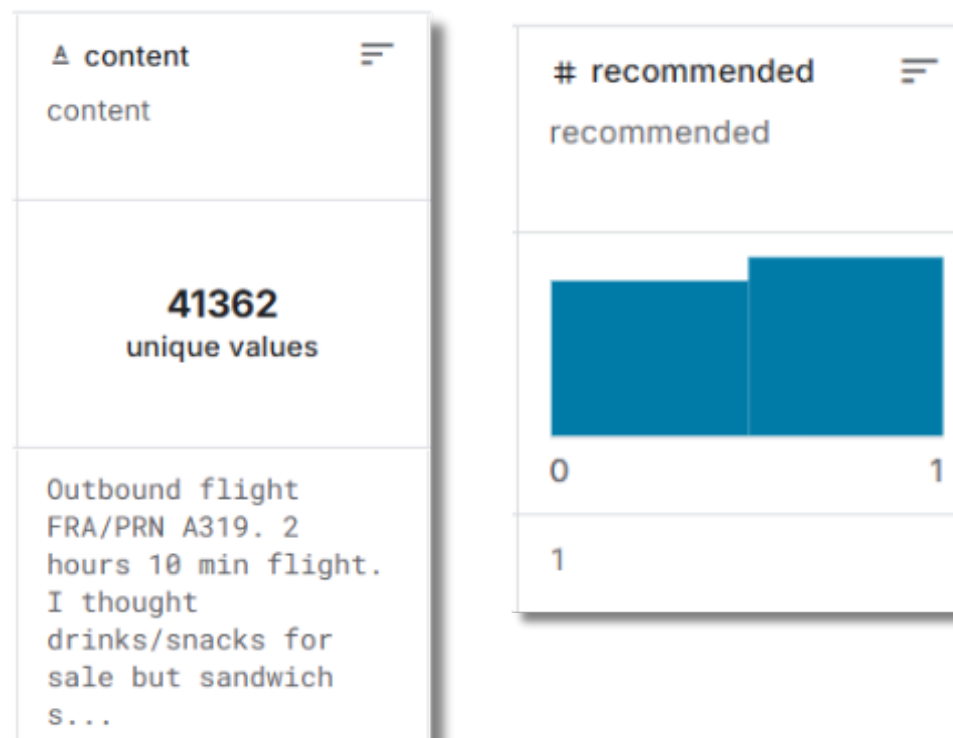
Exploring Data Structures in Sentiment Analysis

Data Tokenization Pre-Processing

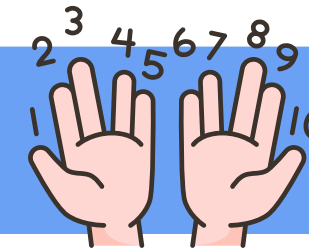


SkyTrax dataset

- Pronoun Filtering
- For a given airline, **split reviews** by their **"recommended"**



Counting the Tokens/Words



- Count Token Frequencies
- Extract Top-K per class

```
[BST] Time (median of 5): 23 ms | Allocated (median): 330 KB
[GOOD] Top 10 most common words in POSITIVE reviews:
1. Word: about      Count: 136
...
[BAD] Top 10 most common words in NEGATIVE reviews:
1. Word: hour       Count: 679
...
```

- Interactive Comparisons

```
=====
COMPARISON RESULTS FOR: "DELAY"
Sentiment: NEGATIVE reviews
Total airlines with this word: 281
Minimum reviews for reliability: 50
=====

TOP 5 Airlines with HIGHEST weighted score for "delay":
(Weighted score considers both frequency % and review count)

Rank Airline      Reviews    Count    Percent    Score
-----
1  sunwing-airlines  286      407     142.31%    349.78
2  allegiant-air     351      375     106.84%    272.07
3  spirit-airlines    786      623     79.26%     229.54
4  norwegian        178      175     98.31%     221.49
5  flybe             227      205     90.31%     212.94

BUILD SUCCESSFUL in 1m 8s
3 actionable tasks: 1 executed, 2 up-to-date
```

Experimental Results



- Analyze Time Complexity and Space Complexity

BST Implementation Test	Red-Black Tree Implementation Test	Arraylist Implementation Test	Map Implementation Test
Time (median of 5): 23 ms Allocated (median): 330 KB	Time (median of 5): 22 ms Allocated (median): 370 KB	Time (median of 5): 53 ms Allocated (median): 2002 KB	Time (median of 5): 23 ms Allocated (median): 3274 KB
[GOOD] Top 10 most common words in POSITIVE reviews:	[GOOD] Top 10 most common words in POSITIVE reviews:	[GOOD] Top 10 most common words in POSITIVE reviews:	[GOOD] Top 10 most common words in POSITIVE reviews:
1. Word: flight Count: 524	1. Word: flight Count: 524	1. Word: flight Count: 524	1. Word: flight Count: 524
2. Word: spirit Count: 330	2. Word: spirit Count: 330	2. Word: spirit Count: 330	2. Word: spirit Count: 330
3. Word: seat Count: 289	3. Word: seat Count: 289	3. Word: seat Count: 289	3. Word: seat Count: 289
4. Word: airline Count: 258	4. Word: airline Count: 258	4. Word: airline Count: 258	4. Word: airline Count: 258
5. Word: time Count: 221	5. Word: time Count: 221	5. Word: time Count: 221	5. Word: time Count: 221
6. Word: all Count: 179	6. Word: all Count: 179	6. Word: all Count: 179	6. Word: all Count: 179
7. Word: fly Count: 153	7. Word: fly Count: 153	7. Word: fly Count: 153	7. Word: fly Count: 153
8. Word: bag Count: 144	8. Word: bag Count: 144	8. Word: bag Count: 144	8. Word: bag Count: 144
9. Word: check Count: 142	9. Word: check Count: 142	9. Word: check Count: 142	9. Word: check Count: 142
10. Word: pai Count: 137	10. Word: pai Count: 137	10. Word: pai Count: 137	10. Word: pai Count: 137
[BAD] Top 10 most common words in NEGATIVE reviews:	[BAD] Top 10 most common words in NEGATIVE reviews:	[BAD] Top 10 most common words in NEGATIVE reviews:	[BAD] Top 10 most common words in NEGATIVE reviews:
1. Word: flight Count: 2117	1. Word: flight Count: 2117	1. Word: flight Count: 2117	1. Word: flight Count: 2117
2. Word: spirit Count: 947	2. Word: spirit Count: 947	2. Word: spirit Count: 947	2. Word: spirit Count: 947
3. Word: airline Count: 904	3. Word: airline Count: 904	3. Word: airline Count: 904	3. Word: airline Count: 904
4. Word: seat Count: 690	4. Word: seat Count: 690	4. Word: seat Count: 690	4. Word: seat Count: 690
5. Word: hour Count: 679	5. Word: hour Count: 679	5. Word: hour Count: 679	5. Word: hour Count: 679
6. Word: delay Count: 623	6. Word: delay Count: 623	6. Word: delay Count: 623	6. Word: delay Count: 623
7. Word: fly Count: 557	7. Word: fly Count: 557	7. Word: fly Count: 557	7. Word: fly Count: 557
8. Word: bag Count: 544	8. Word: bag Count: 544	8. Word: bag Count: 544	8. Word: bag Count: 544
9. Word: time Count: 541	9. Word: time Count: 541	9. Word: time Count: 541	9. Word: time Count: 541
10. Word: get Count: 533	10. Word: get Count: 533	10. Word: get Count: 533	10. Word: get Count: 533

- 5x, 10x, 15x csv to measure asymptotic complexity

```
# --- Configuration ---
# Set the path to your input CSV here. If your
input_csv = "airline.csv" # <-- change this to
# Replication factors you want to output
factors = [10, 100, 1000]
```


Demo



BST Implementation Test	Red-Black Tree Implementation Test	ArrayList Implementation Test	Map Implementation Test
Time (median of 5): 23 ms Allocated (median): 330 KB	Time (median of 5): 22 ms Allocated (median): 378 KB	Time (median of 5): 53 ms Allocated (median): 2602 KB	Time (median of 5): 21 ms Allocated (median): 1274 KB
[GOOD] Top 10 most common words in POSITIVE reviews:	[GOOD] Top 10 most common words in POSITIVE reviews:	[GOOD] Top 10 most common words in POSITIVE reviews:	[GOOD] Top 10 most common words in POSITIVE reviews:
1. Word: flight Count: 524	1. Word: flight Count: 524	1. Word: flight Count: 524	1. Word: flight Count: 524
2. Word: spirit Count: 330	2. Word: spirit Count: 330	2. Word: spirit Count: 330	2. Word: spirit Count: 330
3. Word: seat Count: 289	3. Word: seat Count: 289	3. Word: seat Count: 289	3. Word: seat Count: 289
4. Word: airlin Count: 258	4. Word: airlin Count: 258	4. Word: airlin Count: 258	4. Word: airlin Count: 258
5. Word: time Count: 221	5. Word: time Count: 221	5. Word: time Count: 221	5. Word: time Count: 221
6. Word: all Count: 170	6. Word: all Count: 170	6. Word: all Count: 170	6. Word: all Count: 170
7. Word: fly Count: 153	7. Word: fly Count: 153	7. Word: fly Count: 153	7. Word: fly Count: 153
8. Word: bag Count: 144	8. Word: bag Count: 144	8. Word: bag Count: 144	8. Word: bag Count: 144
9. Word: check Count: 142	9. Word: check Count: 142	9. Word: check Count: 142	9. Word: check Count: 142
10. Word: pai Count: 137	10. Word: pai Count: 137	10. Word: pai Count: 137	10. Word: pai Count: 137
[BAD] Top 10 most common words in NEGATIVE reviews:	[BAD] Top 10 most common words in NEGATIVE reviews:	[BAD] Top 10 most common words in NEGATIVE reviews:	[BAD] Top 10 most common words in NEGATIVE reviews:
1. Word: flight Count: 2117	1. Word: flight Count: 2117	1. Word: flight Count: 2117	1. Word: flight Count: 2117
2. Word: spirit Count: 947	2. Word: spirit Count: 947	2. Word: spirit Count: 947	2. Word: spirit Count: 947
3. Word: airlin Count: 904	3. Word: airlin Count: 904	3. Word: airlin Count: 904	3. Word: airlin Count: 904
4. Word: seat Count: 690	4. Word: seat Count: 690	4. Word: seat Count: 690	4. Word: seat Count: 690
5. Word: hour Count: 679	5. Word: hour Count: 679	5. Word: hour Count: 679	5. Word: hour Count: 679
6. Word: delai Count: 623	6. Word: delai Count: 623	6. Word: delai Count: 623	6. Word: delai Count: 623
7. Word: fly Count: 557	7. Word: fly Count: 557	7. Word: fly Count: 557	7. Word: fly Count: 557
8. Word: bag Count: 544	8. Word: bag Count: 544	8. Word: bag Count: 544	8. Word: bag Count: 544
9. Word: time Count: 541	9. Word: time Count: 541	9. Word: time Count: 541	9. Word: time Count: 541
10. Word: get Count: 533	10. Word: get Count: 533	10. Word: get Count: 533	10. Word: get Count: 533

Array List

Time Complexity:

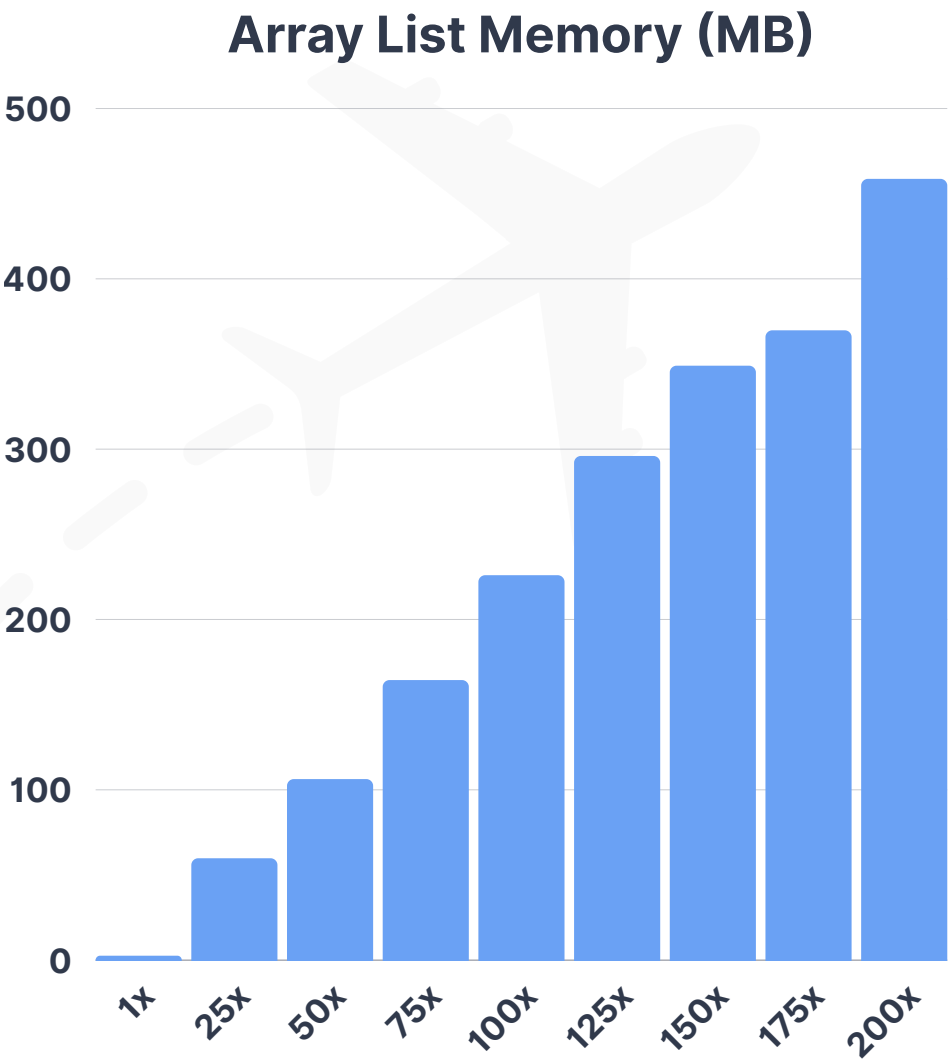
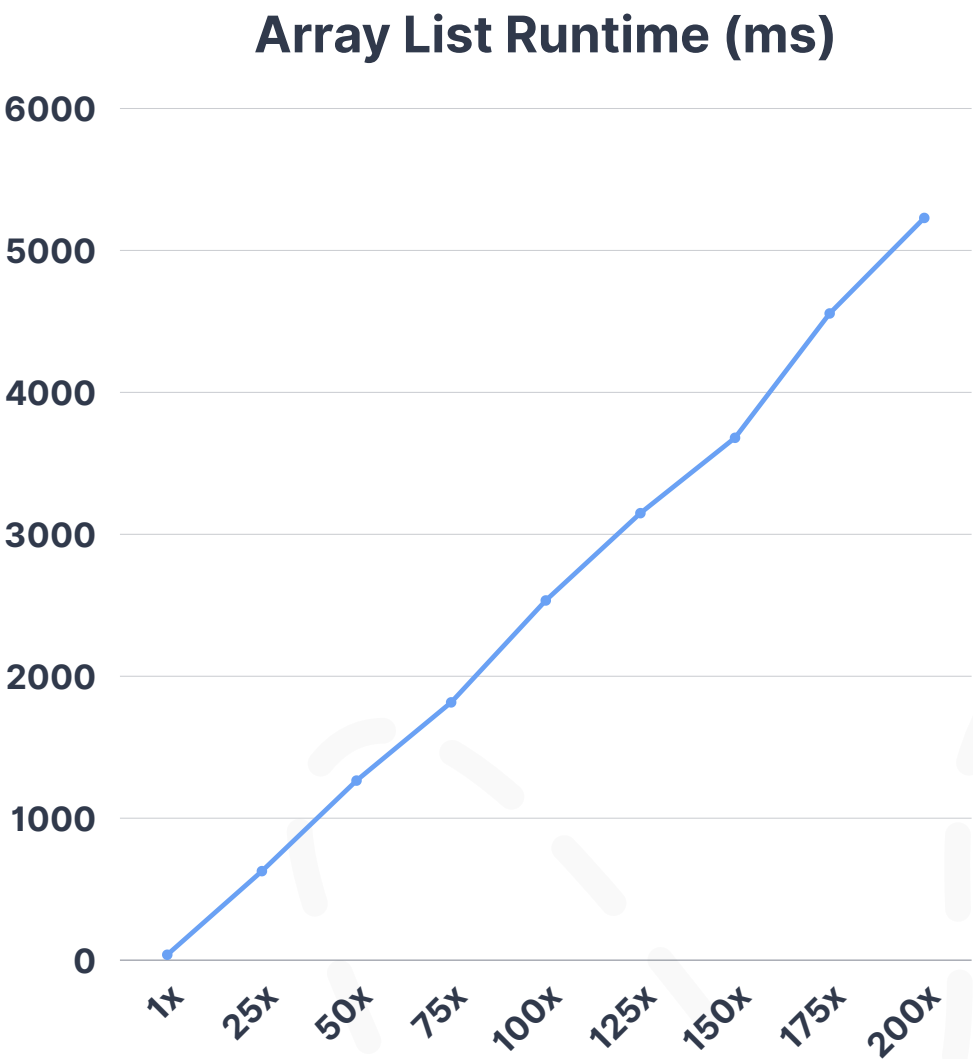
- $O(N \times M \times \log(N \times M))$

Space Complexity:

- $O(N \times M)$

Where:

- **N** = Number of reviews
- **M** = Average words per review



Empirical Results	1x	25x	50x	75x	100x	125x	150x	175x	200x
Runtime (ms)	38	627	1265	1816	2534	3149	3680	4556	5229
Memory (MB)	2.602	59.811	106.294	164.428	225.750	296.118	348.732	369.768	458.685

Map

Time Complexity:

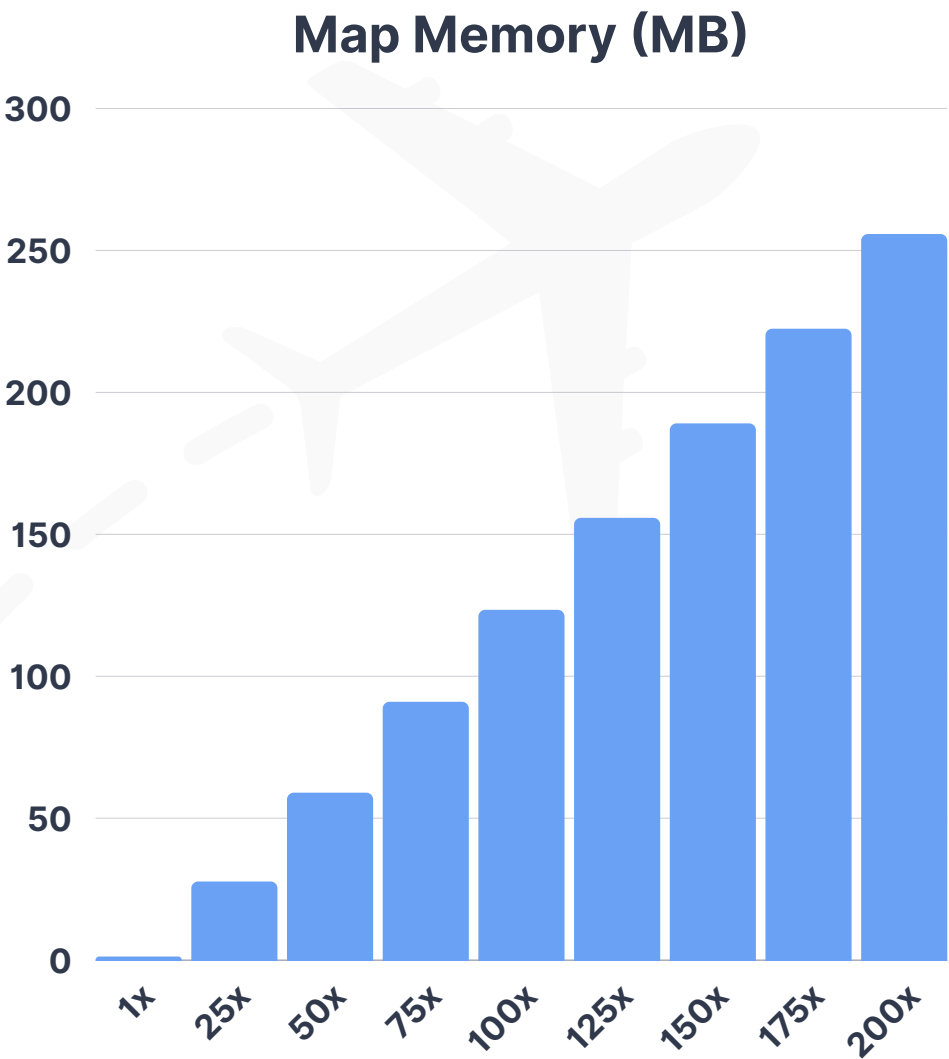
- $O(N \times M + W \log(W))$

Space Complexity:

- $O(W)$

Where:

- **N** = Number of reviews
- **M** = Average words per review
- **W** = Unique words total



Empirical Results	1x	25x	50x	75x	100x	125x	150x	175x	200x
Runtime (ms)	7	86	153	207	271	323	394	453	519
Memory (MB)	1.274	27.701	58.995	91.012	123.409	155.806	189.08	222.429	255.779

Binary Search Tree

Time Complexity:

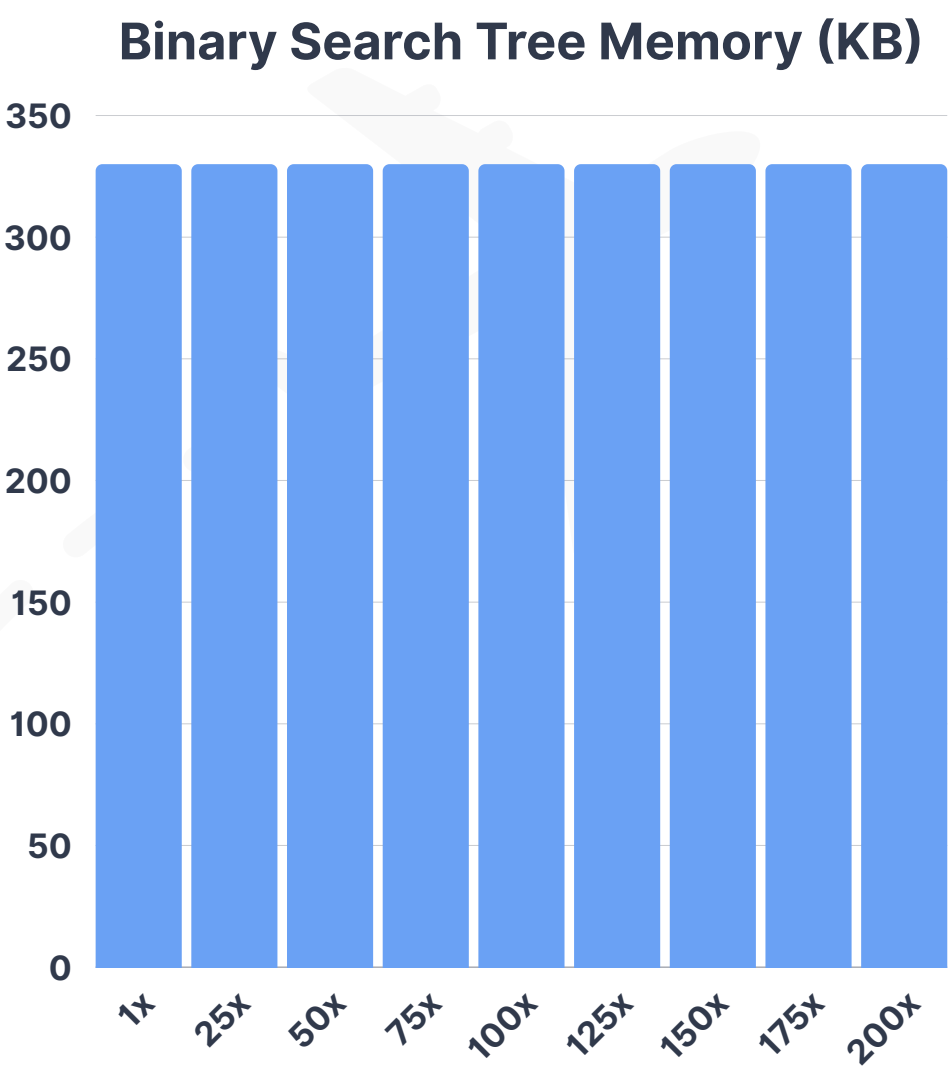
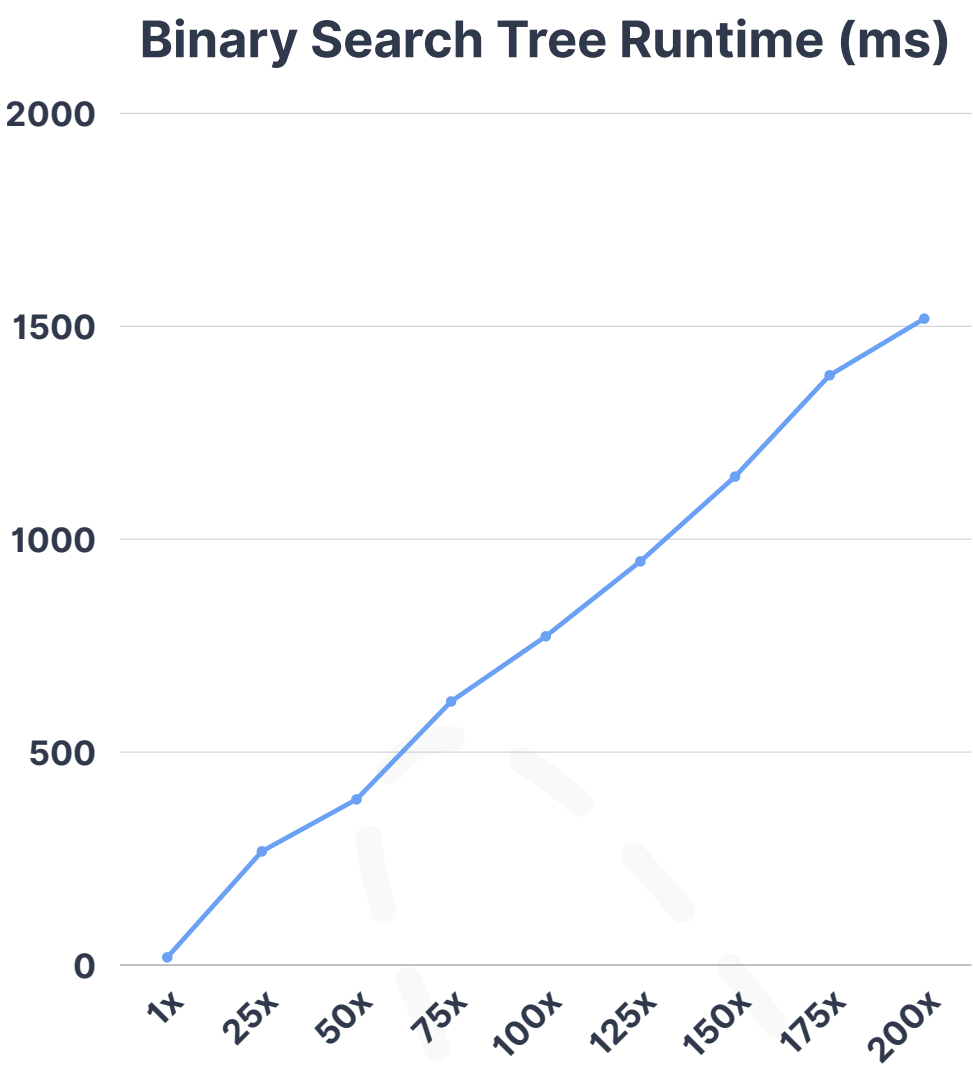
- $O(N \times M \times \log(N \times M))$

Space Complexity:

- $O(N \times M)$

Where:

- **N** = Number of reviews
- **M** = Average words per review



Empirical Results	1x	25x	50x	75x	100x	125x	150x	175x	200x
Runtime (ms)	18	267	389	619	772	948	1147	1385	1518
Memory (MB)	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33

Red-Black Tree

Time Complexity:

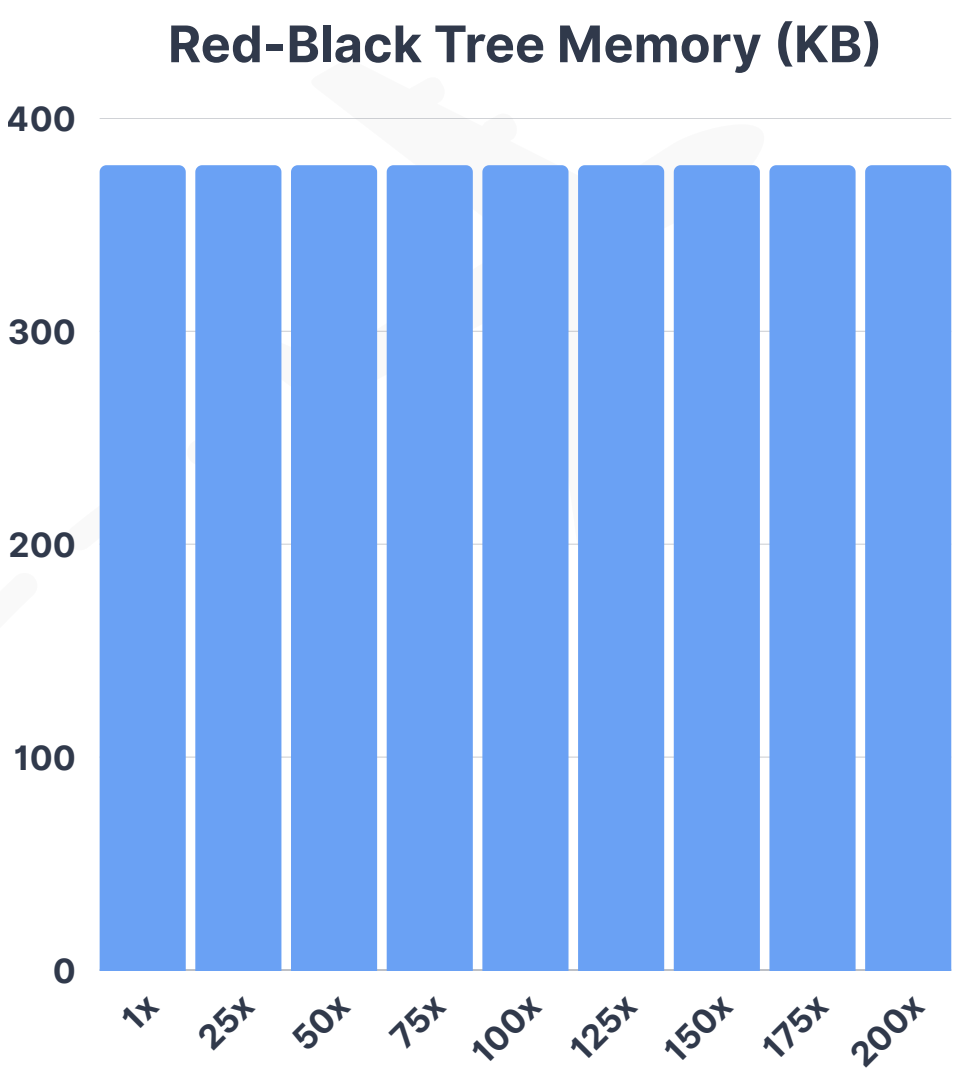
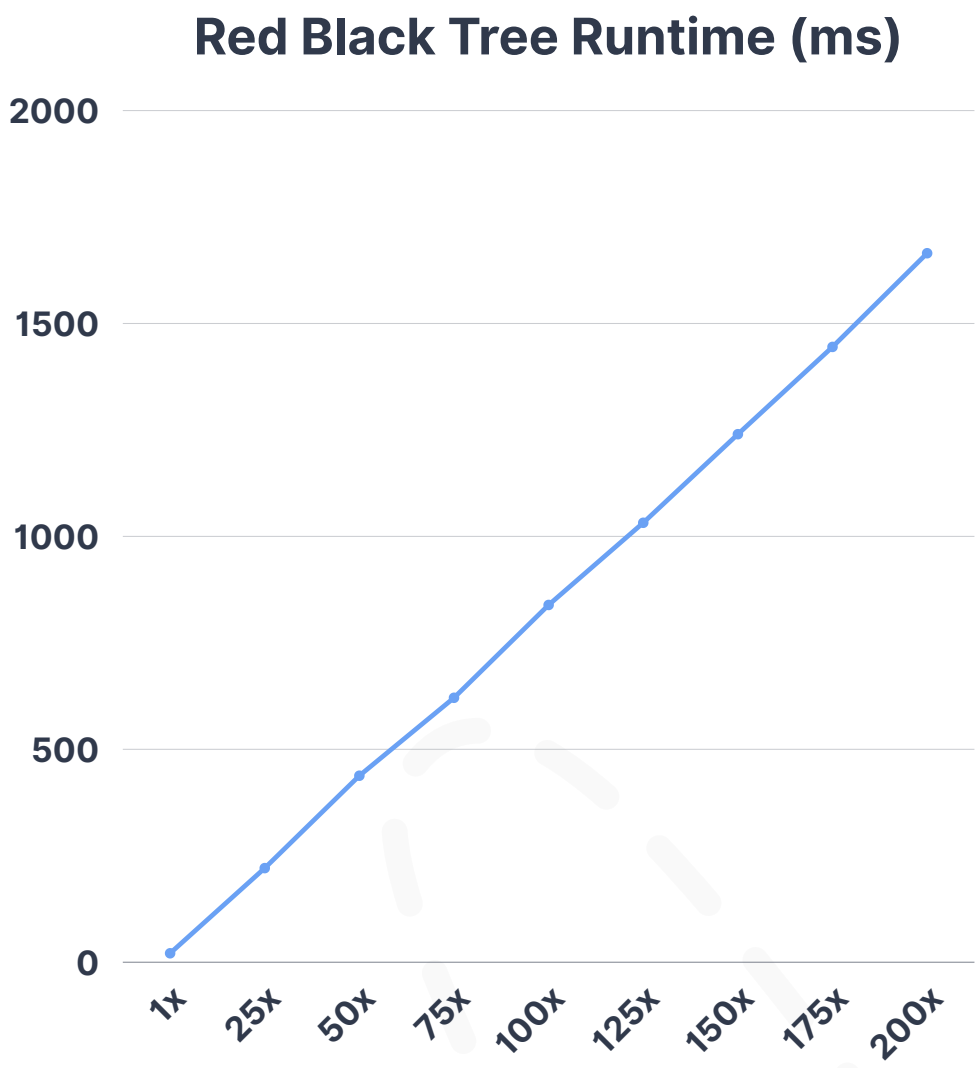
- $O(N \times M \times \log(W))$

Space Complexity:

- $O(W)$

Where:

- **N** = Number of reviews
- **M** = Average words per review
- **W** = Unique Words



Empirical Results	1x	25x	50x	75x	100x	125x	150x	175x	200x
Runtime (ms)	21	221	438	621	839	1032	1240	1445	1665
Memory (MB)	0.378	0.378	0.378	0.378	0.378	0.378	0.378	0.378	0.378

AVL Tree

Time Complexity:

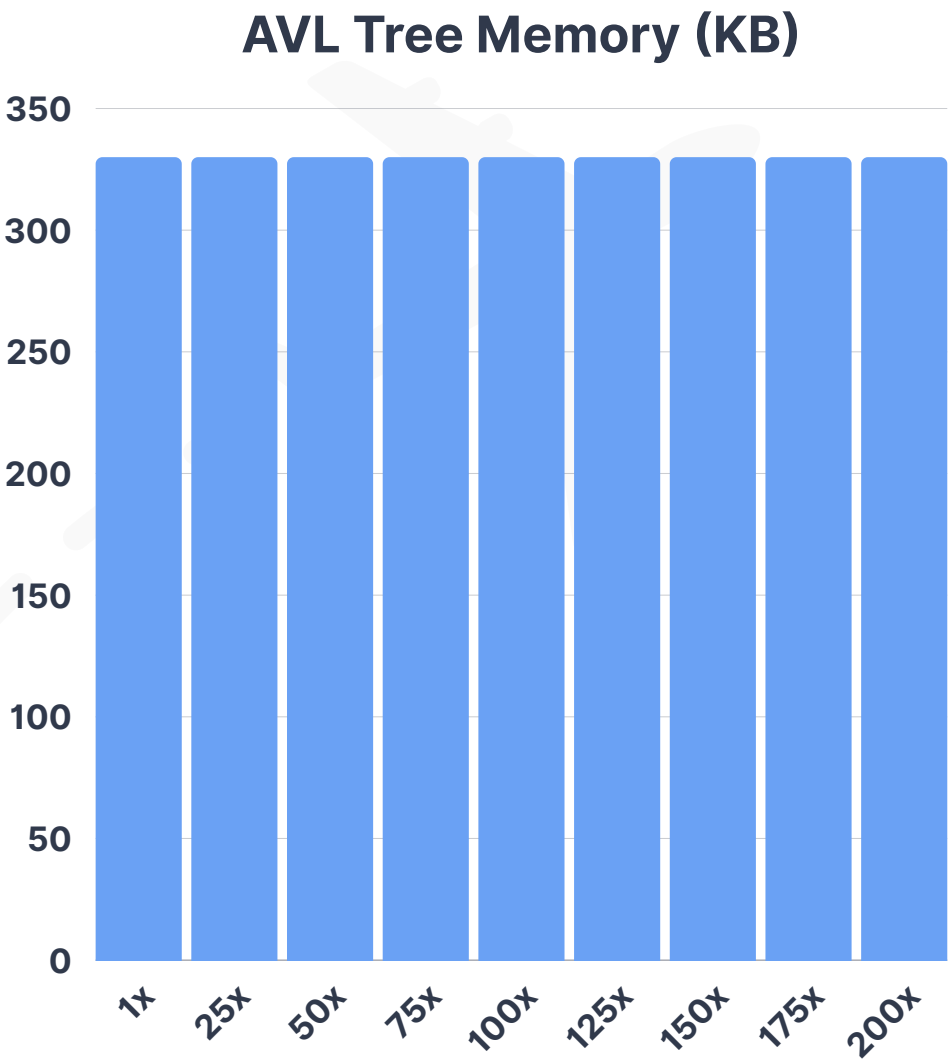
- $O(N \times M \times \log(W))$

Space Complexity:

- $O(W)$

Where:

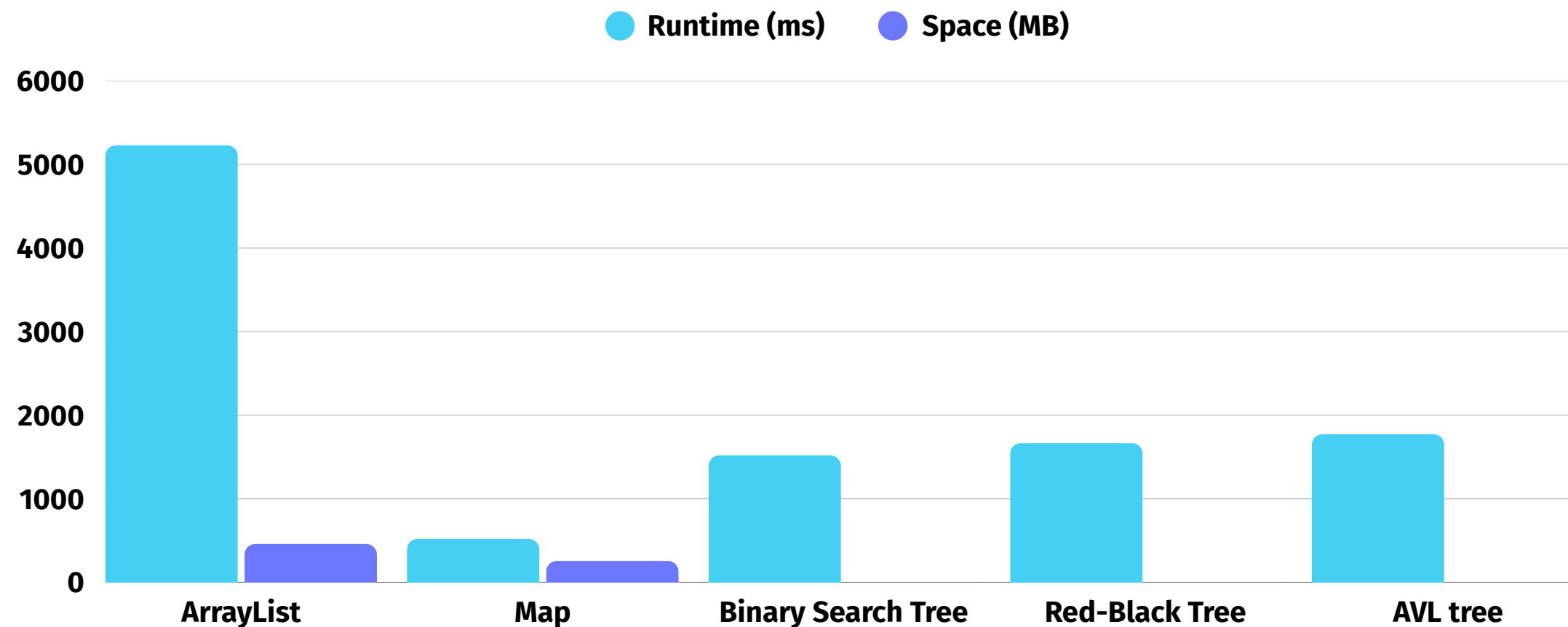
- **N** = Number of reviews
- **M** = Average words per review
- **W** = Unique Words



Empirical Results	1x	25x	50x	75x	100x	125x	150x	175x	200x
Runtime (ms)	26	259	467	681	903	1102	1355	1612	1771
Memory (MB)	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33

Empirical Results: Implementation Advantage

Performance Metrics for Data Structures (ADT)



• For **Top-K** filtering (for 200x dataset)

3 Key Insights



Trees uses least and constant space across datasets



Map is the clear winner on speed (runtime)



Recommendation:
Maps for static reviews, RBT for streaming/live-reviews